

Trace Animation in ExplorVIZ (Demo)

Lauritz Timm

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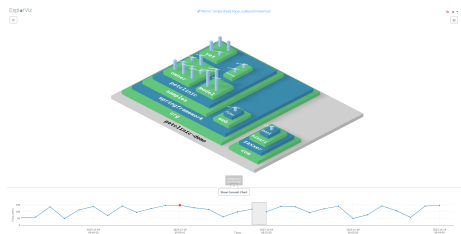
Kiel University
Master Project



Kiel University
Christian-Albrechts-Universität zu Kiel

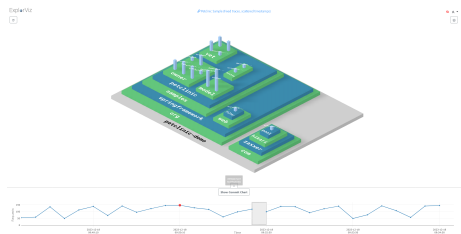
Recap (Challenges)

- Data Points in Time and Space



```
1  "spanList": [  
2    ...  
3    {  
4      "landscapeToken": "7cd8a9a7-b840-4735-9ef0-2dbbfa01c039",  
5      "traceId": "05081bcbd7215fc1c3868df8ee4f69a3",  
6      "spanId": "b5689b0c56e09f8a",  
7      "parentSpanId": "dc838e5090ed9352",  
8      "startTime": 1701953496301,  
9      "endTime": 1701953496301,  
10     "methodHash": "-4433863193898966688"  
11   },  
12   ...  
13 ]
```

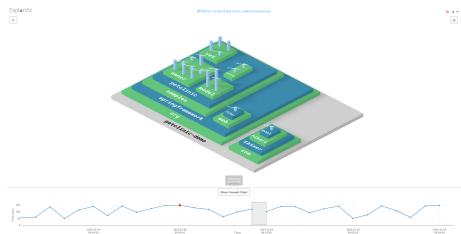
Recap (Challenges)



- Data Points in Time and Space
- Discrete Trace Elements → Continuous Animation

```
1 "spanList": [
2   ...
3   {
4     "landscapeToken": "7cd8a9a7-b840-4735-9ef0-2dbbfa01c039",
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6     "spanId": "b5689b0c56e09f8a",
7     "parentSpanId": "dc838e5090ed9352",
8     "startTime": 1701953496301,
9     "endTime": 1701953496301,
10    "methodHash": "-4433863193898966688"
11  },
12  ...
13 ]
```

Recap (Challenges)



- Data Points in Time and Space
- Discrete Trace Elements → Continuous Animation
- Perception of Transitions / Changes

```
1  "spanList": [  
2    ...  
3    {  
4      "landscapeToken": "7cd8a9a7-b840-4735-9ef0-2dbbfa01c039",  
5      "traceId": "05081bcbd7215fc1c3868df8ee4f69a3",  
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7      "parentSpanId": "dc838e5090ed9352",  
8      "startTime": 1701953496301,  
9      "endTime": 1701953496301,  
10     "methodHash": "-4433863193898966688"  
11   },  
12   ...  
13 ]
```

Prepare Traces

```
1
2 [
3   {
4     "landscapeToken": "7cd8a9a7-b840-4735-9ef0-2dbbfa01c039",
5     "traceId": "05081bcd7215fc1c3868df8ee4f69a3",
6     "spanId": "dc838e5090ed9352",
7     "parentSpanId": "",
8     "startTime": 1701953496300,
9     "endTime": 1701953496315,
10    "methodHash": "276292880809727386"
11  },
12  {
13    "landscapeToken": "7cd8a9a7-b840-4735-9ef0-2dbbfa01c039",
14    "traceId": "05081bcd7215fc1c3868df8ee4f69a3",
15    "spanId": "b5689b0c56e09f8a",
16    "parentSpanId": "dc838e5090ed9352",
17    "startTime": 1701953496301,
18    "endTime": 1701953496301,
19    "methodHash": "-4433863193898966688"
20  },
21  ...
22 ]
```

Prepare Traces

```
1  [
2
3  {
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9    "endTime": 1701953496315,
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15   "spanId": "b5689b0c56e09f8a",
16   "parentSpanId": "dc838e5090ed9352",
17   "startTime": 1701953496301,
18   "endTime": 1701953496301,
19   "methodHash": "-4433863193898966688"
20 },
21 ...
22 ]
```



```
1  [
2    {
3      id: "dc838e5090ed9352",
4      start: 0,
5      end: 15,
6      name: "doFilter(ServletRequest, ServletResponse, FilterChain)",
7      parents: [],
8      children: [
9        {
10          id: "b5689b0c56e09f8a",
11          start: 1,
12          end: 1,
13          name: "setAllowedFields",
14          parents: [
15            {
16              id: "dc838e5090ed9352",
17              ...
18            }
19          ],
20          children: []
21        },
22        ...
23      ]
24    }
25  ]
```

Prepare Traces

```
1 [
2   {
3     {
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21    ...
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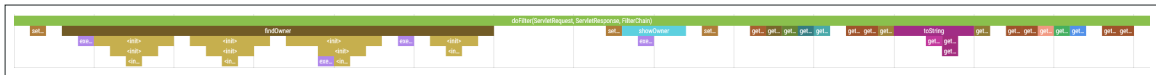
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1 [
2   {
3     id: "dc838e5090ed9352",
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14        parents: [
15          {
16            id: "dc838e5090ed9352",
17            ...
18          }
19        ],
20        children: []
21      },
22      ...
23    ]
24  }
```

Prepare Traces



<https://ui.perfetto.dev/>

Prepare Traces



<https://ui.perfetto.dev/>

Basic Features

- Follow Trace

Basic Features

- Follow Trace

Name	getPets
Call	OncePerRequestFilter → Owner
App	petclinic-demo → petclinic-demo
Time	54 → 55 (1)

Basic Features

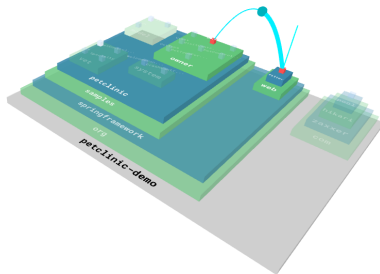
- Follow Trace ✓

Basic Features

- Follow Trace ✓
- Highlighting of Active Elements / Communication

Basic Features

- Follow Trace ✓
- Highlighting of Ad



Basic Features

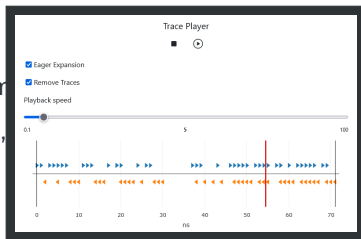
- Follow Trace ✓
- Highlighting of Active Elements / Communication ✓

Basic Features

- Follow Trace ✓
- Highlighting of Active Elements / Communication ✓
- Basic GUI Elements (Start, Stop, Step)

Basic Features

- Follow Trace ✓
- Highlighting of Active Element
- Basic GUI Elements (Start,



Basic Features

- Follow Trace ✓
- Highlighting of Active Elements / Communication ✓
- Basic GUI Elements (Start, Stop, Step Selection, Pause) ✓

Advanced Features

- Transitions

Advanced Features

- Transitions ✓

Advanced Features

- Transitions ✓
- Flow Particles / Visual Elements

Advanced Features

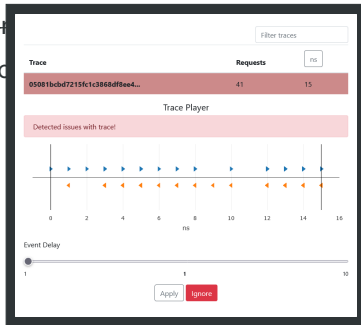
- Transitions ✓
- ~~Flow Particles / Visual Elements~~ Cursor, Trail ✓

Advanced Features

- Transitions ✓
- ~~Flow Particles / Visual Elements~~ Cursor, Trail ✓
- Handle Gaps / Missing Information

Advanced Features

- Transitions ✓
- ~~Flow Particles / Visual Elements~~
- Handle Gaps / Missing Info



Advanced Features

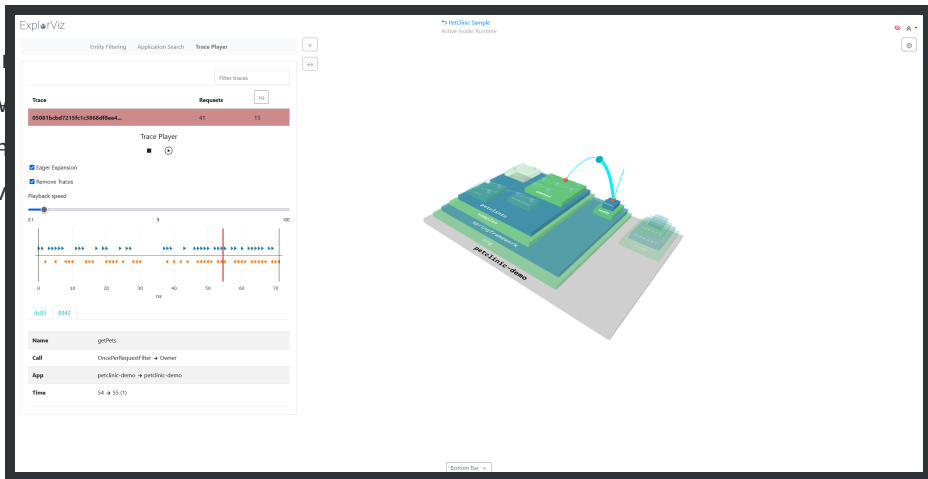
- Transitions ✓
- ~~Flow Particles / Visual Elements~~ Cursor, Trail ✓
- ~~Handle Gaps / Missing Information~~ Trace Tree / Preprocessing ✓

Advanced Features

- Transitions ✓
- ~~Flow Particles / Visual Elements~~ Cursor, Trail ✓
- ~~Handle Gaps / Missing Information~~ Trace Tree / Preprocessing ✓
- Advanced GUI Elements (Interval, Speed, Transition Time, etc.)

Advanced Features

- Trace
- Flow
- Har
- Adv



Advanced Features

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- ~~Flow Particles / Visual Elements~~ Cursor, Trail ✓
- ~~Handle Gaps / Missing Information~~ Trace Tree / Preprocessing ✓
- Advanced GUI Elements (~~Interval~~ Cursor / Selection, Speed, ~~Transition Time~~, etc.) ✓

Advanced Features

- Transitions ✓
- ~~Flow Particles~~ / ~~Visual Elements~~ Cursor, Trail ✓
- ~~Handle Gaps~~ / ~~Missing Information~~ Trace Tree / Preprocessing ✓
- Advanced GUI Elements (~~Interval~~ Cursor / Selection, Speed, ~~Transition Time~~, etc.) ✓
- Unique Color / Height ✓

Advanced Features

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- ~~Flow Particles / Visual Elements~~ Cursor, Trail ✓
- ~~Handle Gaps / Missing Information~~ Trace Tree / Preprocessing ✓
- Advanced GUI Elements (~~Interval~~ Cursor / Selection, Speed, ~~Transition Time~~, etc.) ✓

- Unique Color / Height ✓
- Progress / Timeline ✓

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- Detail Tabs ✓

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- ...